higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

INTRODUCTION TO SYSTEMS DEVELOPMENT
(Second Paper)
NQF LEVEL 2

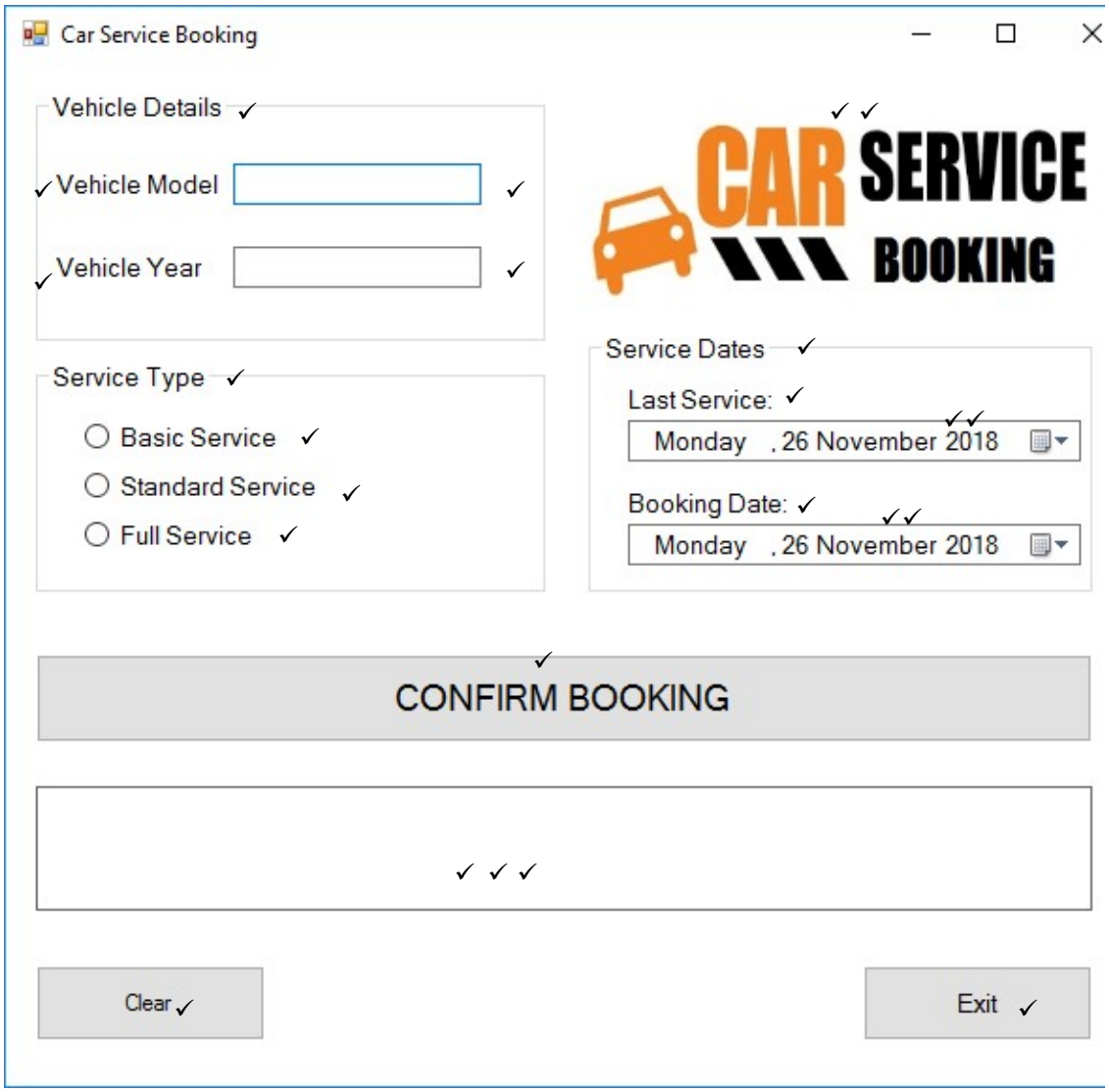
28 NOVEMBER 2019

This marking guideline consists of 7 pages.

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QUESTION 1

1.1 – 1.8



(24)

1.9 Examination Number + Printout of form.✓✓ (2)

[26]

QUESTION 2

2.1
```
Public Class Form1
    Dim decMonthlyIncome As Decimal = 0✓
    Dim decMonthlyExpenses As Decimal = 0✓
```
(2)

2.2.1
```
Dim decSalary As Decimal = 0✓
Dim decOtherIncome As Decimal = 0✓
```
(2)

2.2.2
```
decSalary = CDec(txtSalary.Text)✓
decOtherIncome = CDec(txtOtherIncome.Text)✓
```
(2)

2.2.3
```
decMonthlyIncome = decSalary✓ + decOtherIncome✓
lblTotalIncome.Text = decMonthlyIncome.✓ ToString("C2")✓
```
(4)

2.2.4
```
grpExpenses.Enabled = True✓
```
(1)

2.3 2.3.1
```
Dim decRent As Decimal = 0✓
Dim decGroceries As Decimal = 0✓
Dim decWaterElectricity As Decimal =0✓
Dim decEntertainment As Decimal = 0✓
```
(Any 2 × 1) (2)

2.3.2
```
decRent = CDec(txtRent.Text)✓
decGroceries = CDec(txtGroceries.Text)✓
decWaterElectricity = CDec(txtWaterElec.Text)✓
decEntertainment = CDec(txtEntertainment.Text)✓
```
(4)

2.3.3
```
decMonthlyExpenses = decRent + decGroceries +
decWaterElectricity + decEntertainment✓✓

lblTotalExpenses.Text  =  decMonthlyExpenses.ToString("C2")
✓✓
```
(4)

2.4 2.4.1
```
Dim decNetIncome As Decimal = 0✓
```
(1)

2.4.2
```
decNetIncome = decMonthlyIncome – decMonthlyExpenses✓✓

lblDisplay.Text = "Your net income (loss) for the month is:
" & ✓decNetIncome.ToString("C2")✓
```
(4)

2.5
```
txtSalary.Clear()
txtOtherIncome.Clear()
txtRent.Clear()
txtGroceries.Clear()          ✓
txtWaterElec.Clear()
txtEntertainment.Clear()

lblTotalIncome.Text = ""
lblTotalExpenses.Text = ""    ✓
lblDisplay.Text = ""
grpExpenses.Enabled = False✓
```
(3)

2.6
```
Close()✓
```
(1)
[30]

QUESTION 3

3.1 3.1.1

```
Dim decAmountSpent As Decimal = 0✓
Dim decCardDiscountPercent As Decimal = 0✓
Dim decTotalDiscountPercent As Decimal = 0✓
Dim decLoyaltyDiscountPercent As Decimal = 0✓
Dim decTotalDiscount As Decimal = 0✓
```

(Any 3 × 1) (3)

3.1.2

```
decAmountSpent = CDec(txtSpent.Text)✓
```

(1)

3.1.3

```
If radBronze.Checked = True Then✓
      decCardDiscountPercent = 0.05D✓
ElseIf radSilver.Checked = True Then
      decCardDiscountPercent = 0.07D✓
ElseIf radPlatinum.Checked = True Then
      decCardDiscountPercent = 0.12D✓
End If
```

(4)

3.1.4

```
If chkGreen.Checked = True Then✓
    decLoyaltyDiscountPercent = 0.02D✓
End If
If chkPrime.Checked = True Then
    decLoyaltyDiscountPercent = decLoyaltyDiscountPercent +
0.15D✓✓
End If
```

(4)

3.1.5

```
decTotalDiscountPercent = decCardDiscountPercent +✓
decLoyaltyDiscountPercent✓
```

(2)

3.1.6

```
decTotalDiscount = decAmountSpent✓*✓
decTotalDiscountPercent✓
```

(3)

3.1.7

```
lblDisplay.Text = "Your total discount is:" &✓
ControlChars.NewLine✓ & decTotalDiscount.ToString("C2")✓
```

(3)

3.2

```
radBronze.Checked = False
radSilver.Checked = False      } ✓
radPlatinum.Checked = False

chkGreen.Checked = False       } ✓
chkPrime.Checked = False

txtSpent.Clear()✓
lblDisplay.Text = ""  ✓
```

(4)

3.3

```
Close()✓
```

(1)
[25]

QUESTION 4

4.1-4.2

```
Option Strict On
Option Explicit On

'Examination Number

Public Class Form1

    Private Sub Form1_Load(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MyBase.Load
        'add items to combobox
        cboTeams.Items.Add(2) ✓
        cboTeams.Items.Add(3)
        'select first item
        cboTeams.SelectedIndex = 0
    End Sub

    Private Sub btnCompare_Click(ByVal sender As System.Object, ByVal
e As System.EventArgs) Handles btnCompare.Click
        Dim intGamesT1 As Integer = 0
        Dim intGoalsForT1 As Integer = 0
        Dim intGoalsAgainstT1 As Integer = 0
        Dim intWinsT1 As Integer = 0
        Dim intGoalDifferenceT1 As Integer = 0
        Dim decWinPercentT1 As Decimal = 0

        Dim intGamesT2 As Integer = 0
        Dim intGoalsForT2 As Integer = 0
        Dim intGoalsAgainstT2 As Integer = 0
        Dim intWinsT2 As Integer = 0
        Dim intGoalDifferenceT2 As Integer = 0
        Dim decWinPercentT2 As Decimal = 0

        Dim strWinner As String = ""
        Dim decBestPercent As Decimal = 0

        'assign textbox values to variables
        intGamesT1 = CInt(txtGamesT1.Text) ✓
        intGoalsForT1 = CInt✓ (txtGoalsForT1.Text)
        intGoalsAgainstT1 = CInt(txtGoalsAgainstT1.Text)
        intWinsT1 = CInt(txtWinsT1.Text)

        intGamesT2 = CInt(txtGamesT2.Text)
        intGoalsForT2 = CInt(txtGoalsForT2.Text)
        intGoalsAgainstT2 = CInt(txtGoalsAgainstT2.Text)
        intWinsT2 = CInt(txtWinsT2.Text)
        'calculate goal difference ✓ (make comment)
        intGoalDifferenceT1 = intGoalsForT1-✓✓intGoalsAgainstT1
(change to -)
        intGoalDifferenceT2 = intGoalsForT2 - intGoalsAgainstT2
        'calculate win percentage
        decWinPercentT1 = CDec((intWinsT1 / intGamesT1) * 100)
        decWinPercentT2 = CDec((intWinsT2 / intGamesT2) * 100)
        'determine winner
        If decWinPercentT1 > ✓✓ decWinPercentT2 Then   (change to >)
            strWinner = "Team 1"
            decBestPercent = decWinPercentT1
```

```
        Else
            strWinner = "Team 2"
            decBestPercent = decWinPercentT2
        End If
        'display results
        lstDisplay.Items.Add("Team".PadRight(15) & ✓
"Games".PadRight(10) & "Goals For".PadRight(14) & "Goals
Against".PadRight(15) & "Wins".PadRight(7) & "GD".PadRight(7) & "Win
%")
        lstDisplay.Items.Add("----------------------------------------
----------------------------------------")
        lstDisplay.Items.Add("Team 1".PadRight(15) &
intGamesT1.ToString.PadRight(10) &
intGoalsForT1.ToString.PadRight(14) &
intGoalsAgainstT1.ToString.PadRight(15) &
intWinsT1.ToString.PadRight(7) &
intGoalDifferenceT1.ToString.PadRight(7) & decWinPercentT1.ToString)
        lstDisplay.Items.Add("Team 2".PadRight(15) &
intGamesT2.ToString.PadRight(10) &
intGoalsForT2.ToString.PadRight(14) &
intGoalsAgainstT2.ToString.PadRight(15) &
intWinsT2.ToString.PadRight(7) &
intGoalDifferenceT2.ToString.PadRight(7) & decWinPercentT2.ToString)
        'only add in Team 3 if required
        If grpTeam3.Enabled = True Then
            Dim intGamesT3 As Integer = 0
            Dim intGoalsForT3 As Integer = 0
            Dim intGoalsAgainstT3 As Integer = 0
            Dim intWinsT3 As Integer = 0
            Dim intGoalDifferenceT3 As Integer = 0
            Dim decWinPercentT3 As Decimal = 0

            intGamesT3 = CInt(txtGamesT3.Text)
            intGoalsForT3 = CInt(txtGoalsForT2.Text)
            intGoalsAgainstT3 = CInt(txtGoalsAgainstT3.Text)
            intWinsT3 = CInt(txtWinsT3.Text)
            intGoalDifferenceT3 = intGoalsForT3 - intGoalsAgainstT3
            decWinPercentT3 = CDec((intWinsT3 / intGamesT3) * ✓✓100)
            If decWinPercentT3 > decBestPercent Then
                strWinner = "Team 3"
            End If
            lstDisplay.Items.Add("Team 3".PadRight(15) &
intGamesT3.ToString.PadRight(10) &
intGoalsForT3.ToString.PadRight(14) &
intGoalsAgainstT3.ToString.PadRight(15) &
intWinsT3.ToString.PadRight(7) &
intGoalDifferenceT3.ToString.PadRight(7) & decWinPercentT3.ToString)

        End If
        lstDisplay.Items.Add("")
        lstDisplay.Items.Add✓ ("Based on win percentage, the best
team is " & strWinner & "!")

    End Sub
```

```
    Private Sub cboTeams_SelectedIndexChanged(ByVal sender As
System.Object, ByVal e As System.EventArgs) Handles
cboTeams.SelectedIndexChanged
        'disable Team 3 group box based on combobox selection
        If cboTeams.SelectedIndex = 0 Then
            grpTeam3.Enabled = False
        ElseIf cboTeams.SelectedIndex = 1 Then  ✓✓ (swap True and
False)
            grpTeam3.Enabled = True
        End If ✓
    End Sub

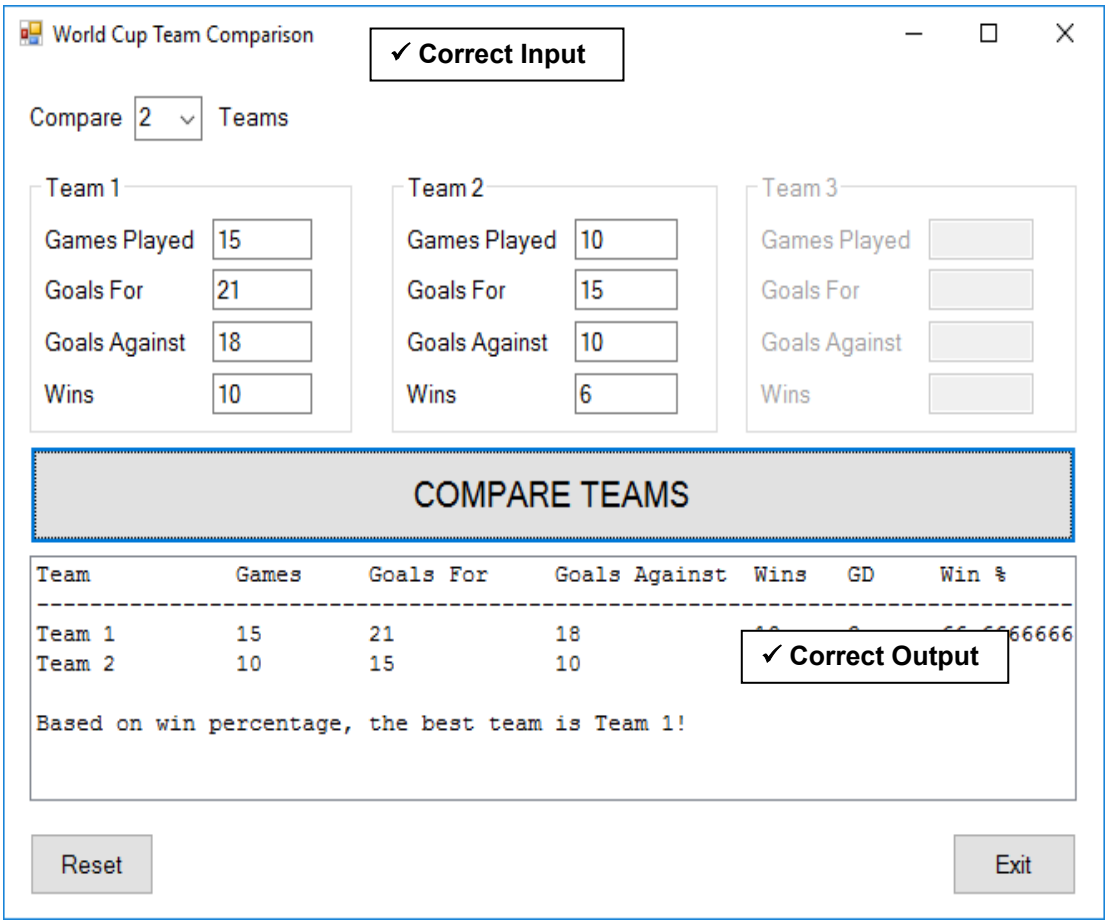
    Private Sub btnReset_Click(ByVal sender As System.Object, ByVal e
As System.EventArgs) Handles btnReset.Click
        'reset controls
        txtGamesT1.Clear()✓
        txtGamesT2.Clear()
        txtGamesT3.Clear()
        txtGoalsAgainstT1.Clear()
        txtGoalsAgainstT2.Clear()
        txtGoalsAgainstT3.Clear()
        txtGoalsForT1.Clear()
        txtGoalsForT2.Clear()
        txtGoalsForT3.Clear()
        txtWinsT1.Clear()
        txtWinsT2.Clear()
        txtWinsT3.Clear()

        lstDisplay.Items.Clear()
        cboTeams.SelectedIndex = 0
        txtGamesT1.Focus()
    End Sub

    Private Sub btnExit_Click(ByVal sender As System.Object, ByVal e
As System.EventArgs) Handles btnExit.Click
        'exit program
        Close()
    End Sub
End Class
```

✓ **Printout of Source Code + Examination Number** (17)

4.3



(2)
[19]

TOTAL: 100

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